

Sales Forecasting & Demand Intelligence System

Executive Summary Report

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Executive Summary

This project analyzes four years of Superstore sales data to understand sales trends, forecast future demand, detect unusual sales patterns, and group products based on demand behavior. Different forecasting models were tested, and XGBoost produced the best performance. The insights from this project can help businesses improve inventory planning and demand forecasting.

Key Findings

- **Revenue by category:** Technology is the top revenue-generating category (₹8,27,455.87 total), ahead of Furniture and Office Supplies.
- **Regional growth:** The West region shows the most consistent sales growth across all four years and posted the highest sales of any region in 2018.
- **Shipping performance:** Average shipping time is about 4 days company-wide; the Central region ships slowest, East ships fastest.
- **Seasonality:** Sales consistently spike in September, November, and December every year — a reliable, repeating pattern rather than one-off events.
- **Best forecasting model:** Among SARIMA, Prophet and XGBoost, the XGBoost model achieved the lowest prediction error and was selected as the final forecasting model (MAE ₹9,812) and the lowest percentage error (MAPE 11.2%) — XGBoost achieved the lowest MAE, RMSE and MAPE among the three models and was selected for forecasting.

Three-Month Sales Forecast

Using the XGBoost model, projected total sales for the next three months are shown below. The model predicts a gradual increase in sales over the next three months, indicating a positive sales trend.

Month	Forecasted Sales	Expected Range
January 2019	₹54,889	₹40,900 – ₹68,900
February 2019	₹56,255	₹42,100 – ₹70,400
March 2019	₹62,240	₹48,100 – ₹76,400

The forecast shows an upward trend heading into Q1 2019, consistent with the year-over-year growth pattern seen in the historical data. Among the segments modeled individually, the East region and Office Supplies category show the highest forecasted sales for Q1 2019, even though Technology remains the strongest category historically.

These forecasts can help the supply chain team plan inventory levels, reduce stock-outs, and avoid unnecessary overstocking.

Top Anomalies Detected

An automated anomaly detector (Isolation Forest) flagged 11 unusual sales weeks out of four years of data. The three most extreme are:

- **March 2015 spike:** Sales spiked to ₹37,704 for the week of March 22, 2015 — far above the typical weekly range at that point in the business. This may have been caused by bulk purchases, seasonal promotions, or special business events.
- **December 2018 spike:** Sales reached ₹35,999 for the week of December 2, 2018 — consistent with early holiday-season buying.
- **November 2018 spike:** Sales hit ₹29,017–₹30,572 across two consecutive weeks in November 2018 — consistent with a festive season or promotional offers like Black Friday / Cyber Monday-style promotional period.

A second method (Z-score on rolling averages) did not flag any of these same weeks at the standard threshold, meaning these spikes are large in absolute terms but happen gradually enough that they don't look sudden relative to the recent trend. The Isolation Forest detected more anomalies than the Z-score method. This suggests that Isolation Forest is better at identifying unusual sales patterns in this dataset.

Product Demand Segmentation

Products were grouped into three demand clusters based on sales volume, order value, volatility, and growth rate:

- **Low Volume, Stable Demand** — Art, Bookcases, Envelopes, Fasteners, Furnishings, Labels, Supplies. Recommendation: keep moderate stock, replenish on demand — no need to over-invest in inventory here.
- **High Volume, Stable Demand** — Accessories, Binders, Chairs, Copiers, Machines, Phones, Storage. Suggested inventory strategy: these are core revenue drivers — maintain high stock availability and avoid stockouts.
- **Growing Demand** — Appliances, Paper, Tables. Recommendation: increase inventory gradually and monitor closely; demand is rising and under-stocking risks lost sales.

Business Recommendations

- 1. Prioritize inventory for Office Supplies and the East region.** Office Supplies has the strongest projected Q1 2019 forecast (₹80,913 combined), and the East region leads regionally (₹80,883 combined) — while Technology remains the top revenue driver historically and shouldn't be deprioritized.
- 2. Build seasonal stock ahead of Sept–Dec.** Sales reliably spike in Sep/Nov/Dec every year, and the two largest anomalies this dataset detected both fell in Nov–Dec 2018 — Maintain additional inventory before September–December because sales consistently increase during these months.
- 3. Monitor the Growing Demand cluster closely.** Appliances, Paper, and Tables show the fastest year-over-year growth rates of any sub-category — Monitor Appliances, Paper and Tables carefully because demand is increasing.

Risk & Limitation

This forecasting system is built on four years of internal sales history only — it does not account for external factors such as pricing changes, marketing promotions, competitor activity, or macroeconomic shifts. The XGBoost model's ₹9,800 average error also reflects a relatively short training window (under four years of monthly data), so forecast confidence should be treated as directional rather than precise, and the model should be retrained regularly as new sales data becomes available.